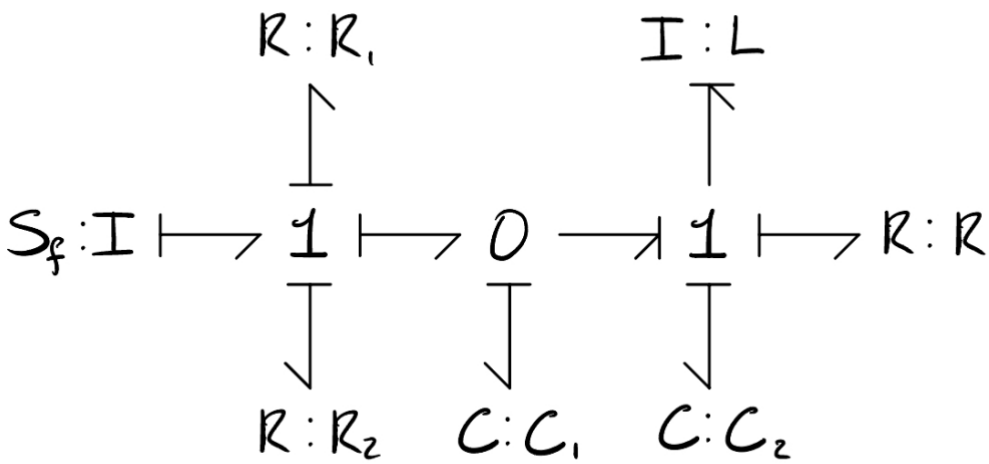
Homework #2 Solutions

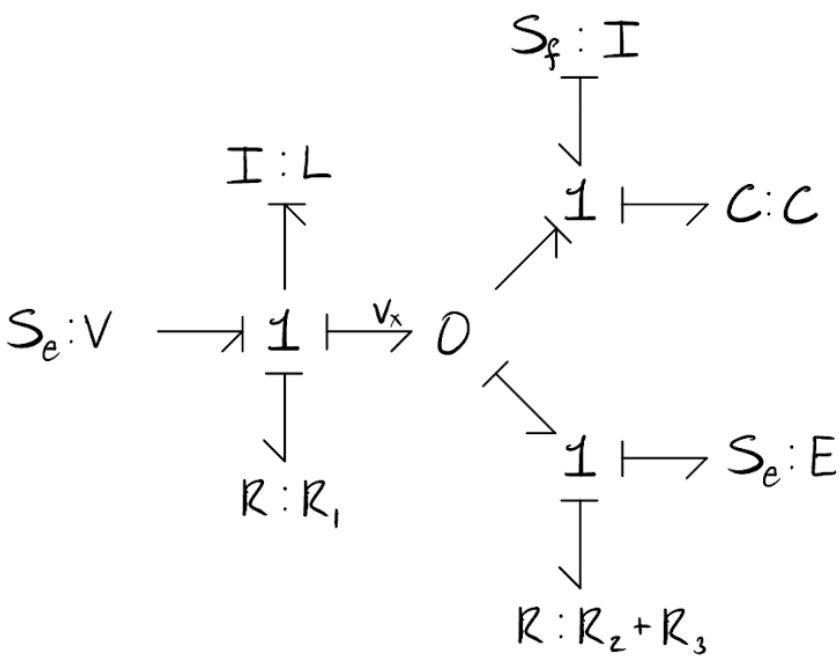
10 points per problem.

- One bond is worth 0.5 points.
- Causal stroke is work 0.5 points.

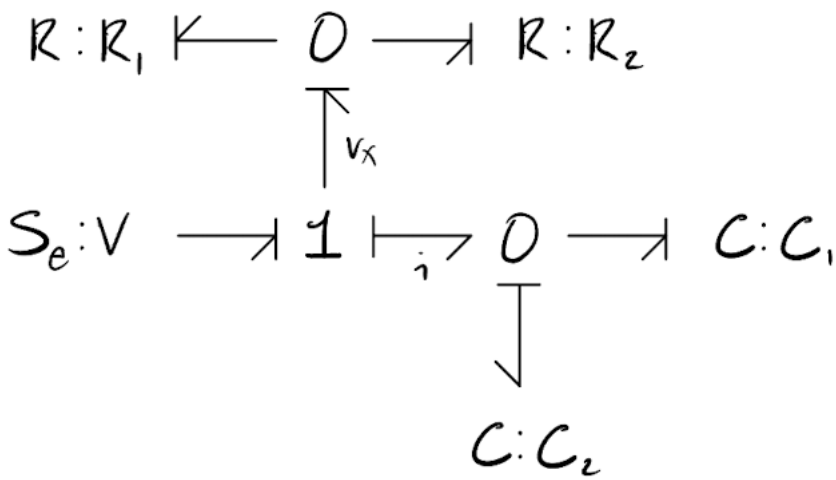
1)



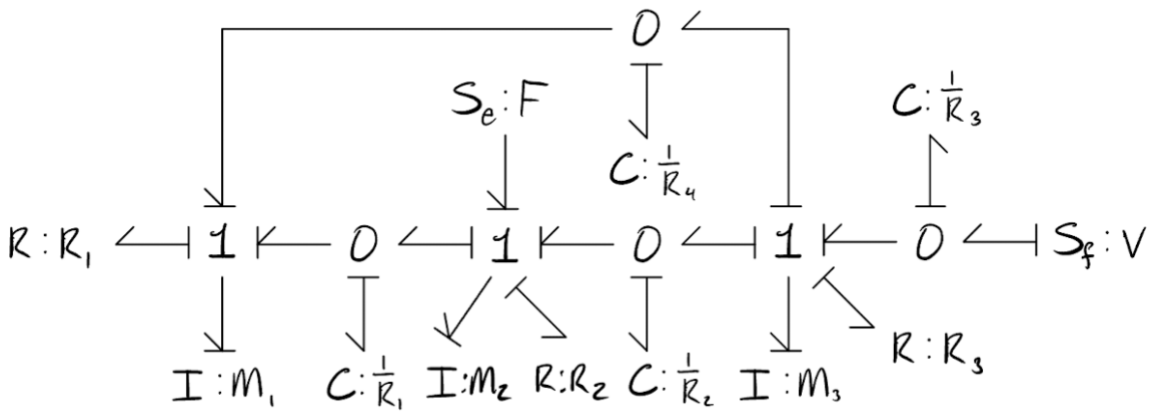
2)



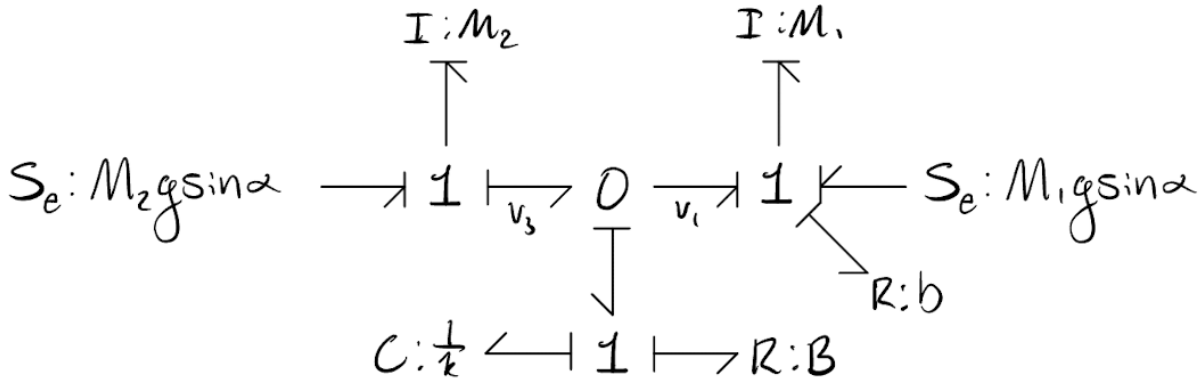
3)


R : R1
0
R : R2
vx
Se : V
1
0
C : C1
C : C2

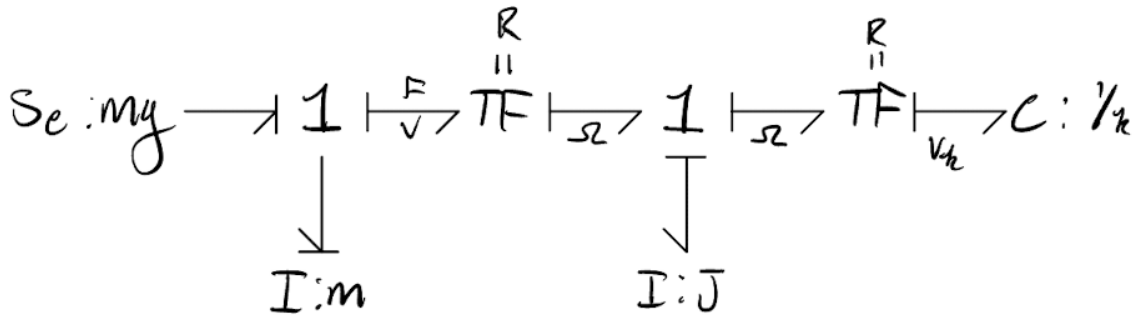
4)


0
Se : F
C : 1/R4
C : 1/R3
R : R1
1
0
1
0
1
0
Sf : V
I : M1
C : 1/R1
I : M2
R : R2
C : 1/R2
I : M3
R : R3

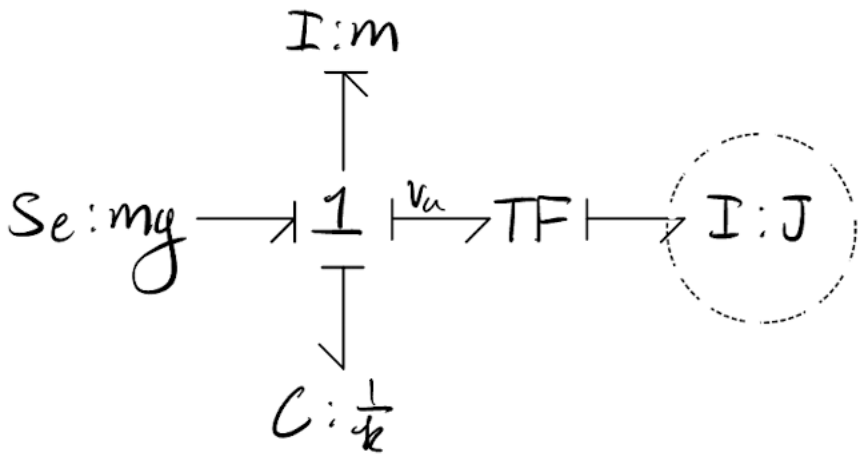
5)


I : M2
I : M1
Se : M2gsinα
1
0
1
Se : M1gsinα
v3
v1
R : b
C : 1/k
1
R : B

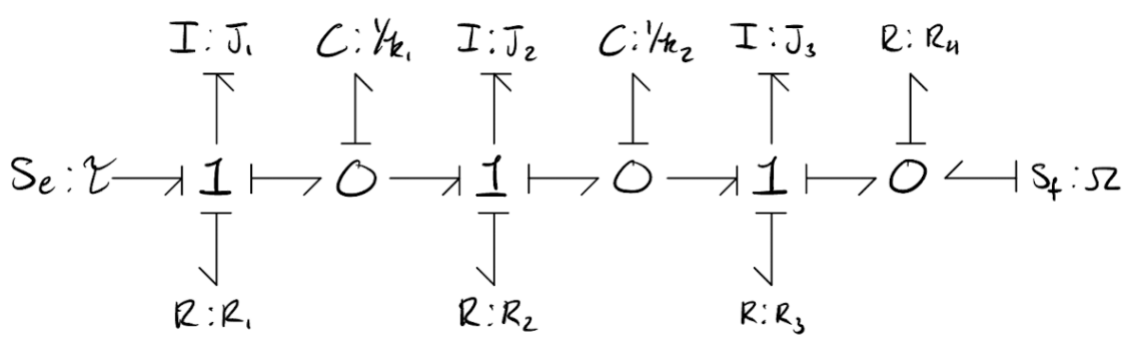
6)


Se : mg
1
F
v
R
TF
Ω
1
Ω
R
TF
v_k
C : 1/k
I : m
I : J

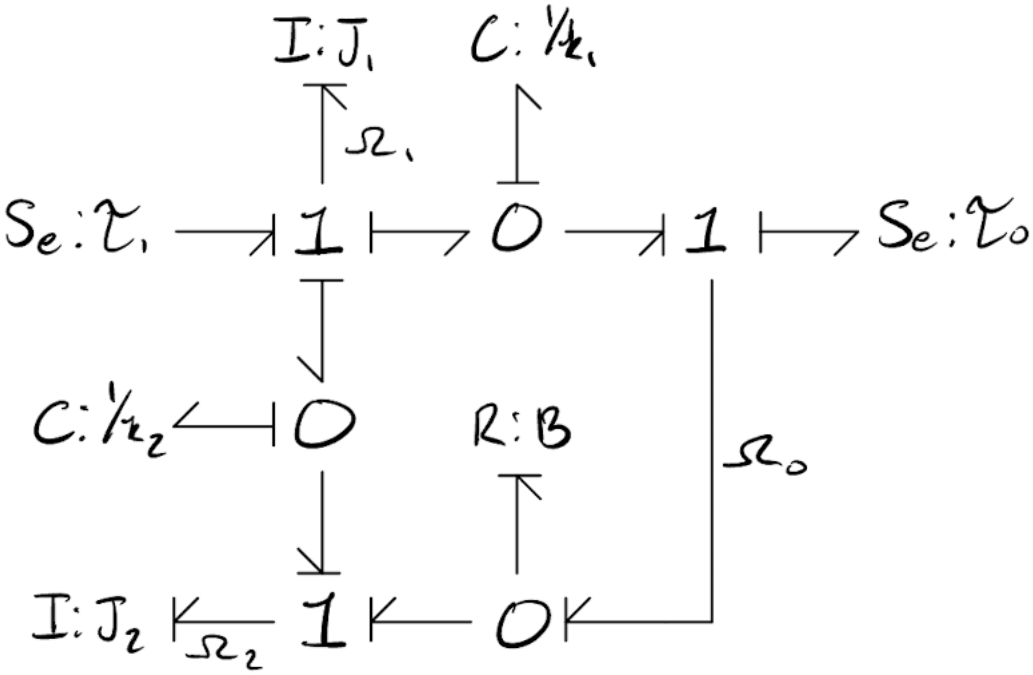
OR


I : m
Se : mg
1
v_u
TF
I : J
C : 1/k

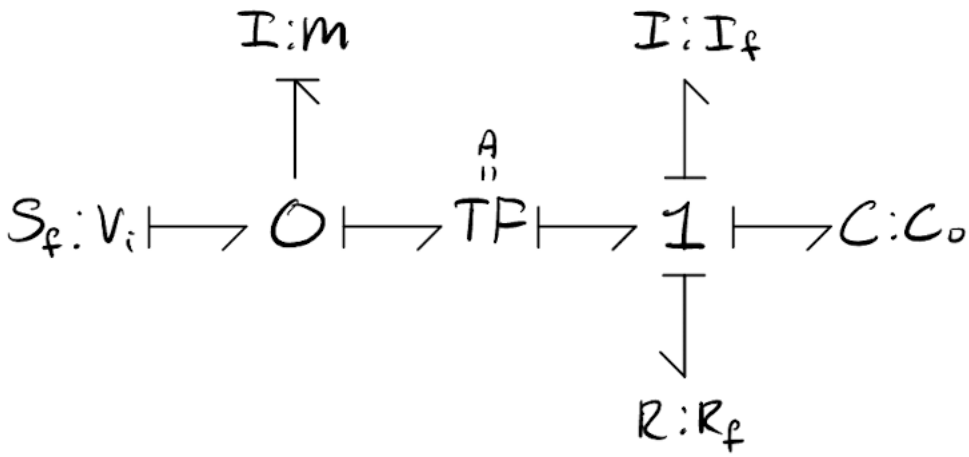
7)


I : J_1
C : $1/k_1$
I : J_2
C : $1/k_2$
I : J_3
R : R_4
Se : τ
1
0
1
0
1
0
Sf : Ω
R : R_1
R : R_2
R : R_3

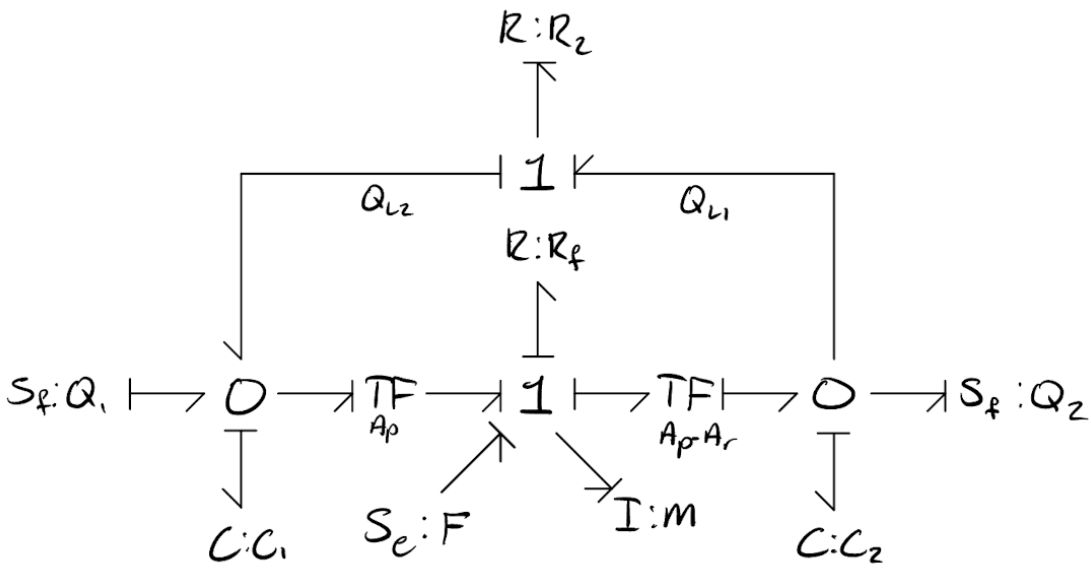
8)



9)


I:m
I:I_f
A
S_f:V_i
0
TF
1
C:C_0
R:R_f

10)


R:R_2
1
Q_L2
Q_L1
R:R_f
S_f:Q_1
0
TF
A_p
1
TF
A_p·A_r
0
S_f:Q_2
C:C_1
S_e:F
I:m
C:C_2